

Program 5

Implement a stack using arrays

```
#include <stdio.h>

#define MAXSIZE 8

int stack[MAXSIZE];

int top = -1;

int isempty() {
    return (top == -1);
}

int isfull() {
    return (top == MAXSIZE - 1); // corrected condition
}

int peek() {
    return stack[top];
}

void pop() {
    if (!isempty()) {
        int data = stack[top];
        top--;
        printf("Popped element: %d\n", data);
    } else {
        printf("Stack is empty! Cannot pop.\n");
    }
}
```

```

void push(int data) {
    if (!isfull()) {
        top++;
        stack[top] = data;
        printf("Pushed element: %d\n", data);
    } else {
        printf("Stack is full! Cannot push.\n");
    }
}

void display() {
    if (isempty()) {
        printf("Stack is empty!\n");
        return;
    }
    printf("Stack elements (top to bottom):\n");
    for (int i = top; i >= 0; i--) {
        printf("%d\n", stack[i]);
    }
}

int main() {
    int choice, value;
    while (1) {
        printf("\n--- Stack Menu ---\n");
        printf("1. Push\n");
        printf("2. Pop\n");
    }
}

```

```
printf("3. Peek\n");
printf("4. Display\n");
printf("5. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice) {
case 1:
    printf("Enter value to push: ");
    scanf("%d", &value);
    push(value);
    break;
case 2:
    pop();
    break;
case 3:
    if (!isempty())
        printf("Top element: %d\n", peek());
    else
        printf("Stack is empty!\n");
    break;
case 4:
    display();
    break;
case 5:
    printf("Exiting...\n");
    return 0;
```

default:

```
printf("Invalid choice! Please try again.\n");
```

```
}
```

```
}
```

```
}
```